

**PART-1 : MATHEMATICS****SECTION-I (i)**

1) The number of subsets  $R$  of  $P = \{1, 2, 3, \dots, 9\}$  which satisfies the property. "There exist integers  $a < b < c$  with  $a \in R, b \notin R, c \in R$ " is

- (A) 512
- (B) 466
- (C) 467
- (D) None of these

2) Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function satisfying  $f(x).f(y) - f(xy) = x + y$  for all  $x, y \in \mathbb{R}$ . The value of  $f(31)$  is

- (A) 32
- (B) -32
- (C) 16
- (D) -16

3) The value of  $\frac{3}{2}\cos^{-1}\left(\frac{2}{\sqrt{7}}\right) + \frac{1}{2}\sin^{-1}\left(\frac{4\sqrt{3}}{7}\right) + \frac{5}{6}\tan^{-1}\left(\frac{9\sqrt{3}}{10}\right)$  is

- (A)  $\frac{4\pi}{3}$
- (B)  $\frac{5\pi}{6}$
- (C)  $\frac{7\pi}{6}$
- (D)  $\frac{2\pi}{3}$

4) If  $A = \{1, 2, 3, 4, 5\}$  then number of functions  $f : A \rightarrow A$  such that  $f(f(x)) = x \forall x \in A$

- (A) 26
- (B) 28
- (C) 20
- (D) 22

**SECTION-I (ii)**

1) Let  $b > a > 0$  and a function  $f$  be defined  $f(x) = \min\{x^2 - 2ax + 3a^2, x^2 - 2bx + b^2 + 2a^2\}$ . If minimum value of  $f(x)$  is 1 then identify which of the following statement(s) is(are) correct

(A) If  $f(x) = \frac{3}{2}$  has 3 solutions then  $b = \sqrt{2}$

(B) If  $f(x) = \frac{3}{2}$  has 3 solutions then  
 $b = \frac{3}{\sqrt{2}}$

(C) Least integral value of  $b$  for which  $f(|x|) = \frac{3}{2}$  has 7 solutions is 2

(D) Least integral value of  $b$  for which  $f(|x|) = \frac{3}{2}$  has 7 solutions is 3

2) Consider a function  $f(x) = [n(\tan^{-1}\{x\} - \cot^{-1}[x])]$ . Identify which of the following option(s) is(are) correct ?

[Note :  $[k]$  and  $\{k\}$  denote greatest integer function and fractional part function of  $k$  respectively]

(A) Number of integers in the domain of  $f(x)$  is 0.

(B) Number of positive integers in the range of  $f(x)$  is 0

(C)  $f(x)$  is not defined for all  $x \in \left(n, n + \frac{1}{n}\right) \forall n \in \mathbb{N}$

(D)  $f(x)$  is defined for all  $x \in \left(n + \frac{1}{n}, n + 1\right) \forall n \in \mathbb{N}, n \geq 2$

3) If  $n(A) = p$ ,  $n(B) = q$  and  $n(A \cap B) = r$ , then identify which of the following statement is(are) correct ?

(A)  $n((A \times B) \cap (B \times A)) = r^2$

(B)  $n((A \times B) \cup (B \times A)) = 2pq - r^2$

(C)  $n((A \times A) \cap (B \times B)) = r^2$

(D)  $n((A \times A) \cup (B \times B)) = p^2 + q^2 + r^2$

4) Consider,  $f(x) = [x]$ .  $\{x\}$  where  $[k]$  and  $\{k\}$  denote greatest integer function and fractional part function of  $k$  respectively. If  $g : \mathbb{R} \rightarrow \mathbb{R}$  is another function such that  $g(x) = f(x) - f(-x)$ . Then

(A) The sum of all the integers in the range of  $g(x)$  is 0

(B) The sum of all the integers in the range of  $g(x)$  is 1

(C) The sum of all the solutions of the equation  $g(x) = x$  in  $(-5, 10)$ , is  $\frac{75}{2}$

(D) The sum of all the solutions of the equation  $g(x) = x$  in  $(-5, 10)$ , is  $\frac{65}{2}$

5) Let  $n$  be the number of integers in the range of  $f(x) = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$ . If a set  $A = \{1, 2, 3, \dots, n\}$  and a relation  $R$  is defined on  $A$  such that  $aRb \Leftrightarrow a + b$  is a perfect square. Then

(A) The number of ordered pairs  $(a, b)$  in  $R$  is 9

(B) The number of ordered pairs  $(a, b)$  in  $R$  is 8

(C) Least number of ordered pairs to be added in  $R$  so that it becomes transitive is 9

(D) Least number of ordered pairs to be added in R so that it becomes transitive is 8

6) Let  $S_n = \sum_{r=1}^n \left( \tan^{-1} \left( \frac{1}{r^2 + r + 1} \right) + \tan^{-1} \left( \frac{2r}{r^4 + r^2 + 2} \right) \right) + \tan^{-1} \left( \frac{4}{n^2 + 4n + 1} \right)$ . The smallest natural number n for which  $\tan(S_n) > 2030$  is

- (A) 2021
- (B) 2027
- (C) 2023
- (D) 2022

### SECTION-III

1) Number of integers satisfying  $|x^2 - 5x + 6| + |x^2 + x + 1| = |6x - 5|$  is equal to

2) The number of positive integers n such that  $3n - 4$ ,  $4n - 5$ ,  $5n - 3$  are all prime integers is equal to

3) Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function such that  $f(x) f(x + 2) = -1$  for all  $x \in \mathbb{R}$ . If  $f(1001) = 2$ , then the value

of  $\frac{1}{11} \left( \sum_{r=1}^{101} f(2r-1) \right)$  is

4) Let  $A = \{1, 2, 3, 4, 5\}$ . If R is an equivalence relation on set A such that  $(1, 2) \in R$ ,  $(3, 4) \in R$  but  $(2, 3) \notin R$  then number of relations R is

5) If  $S = \left\{ x \in \mathbb{R} : \sin^{-1} \left( \frac{x+1}{x^2+2x+2} \right) - \sin^{-1} \left( \frac{x-1}{x^2-2x+2} \right) = \frac{\pi}{4} \right\}$  then  $\sum_{x \in S} \left( \sin \left( \sin(x^2+3) \frac{\pi}{2} \right) - \cos \left( (x^2+3) \pi \right) + \tan^{-1} x \right)$  is equal to

6) If  $\alpha$  and  $\beta$  are the smallest and greatest real number of p in  $\left[ 0, \frac{5\pi}{2} \right]$  such that  $\sin^{-1}(\sin p) - \sin^{-1}(\cos p) = -\frac{\pi}{6}$ , then the value of  $\left[ \frac{3(\alpha + \beta)}{7} \right]$  is  
[Note :  $[k]$  denotes greatest integer less than or equal to k]

7) The number of solution(s) of the equation  $\tan^{-1}|x^2 - 2x| + \cot^{-1}(x^2 - 2x + 2) = \frac{\pi}{4}$  is

8) Let  $A = \{1, 2, 3, 4, 5\}$ . If number of symmetric but not reflexive relations on set A such that  $(2, 3) \in R$  but  $(2, 4) \notin R$  is  $m \cdot 2^n$  where m & n coprime then  $(m - 3n)$  is

## PART-2 : PHYSICS

### SECTION-I (i)

1) A point particle P is projected horizontally from a point 5m above a horizontal plane with speed  $10\text{m s}^{-1}$ . It subsequently moves under gravity. A stationary plane mirror is placed vertically such that its plane makes an angle  $45^\circ$  vertical plane containing the trajectory of the particle. At  $t = 0$ , the perpendicular distance of the particle from the mirror is  $\frac{5}{\sqrt{2}}\text{m}$ . Taking the initial position of the image as the origin, with the X-axis horizontal along the initial velocity of the image and the Y-axis vertically upward, the trajectory of the image is

- (A)  $Y = -\frac{X^2}{20}$   
 (B)  $Y = -\frac{X^2}{10}$   
 (C)  $Y = -\frac{X^2}{10\cos^2 45^\circ}$   
 (D)  $Y = -\frac{X^2}{40}$

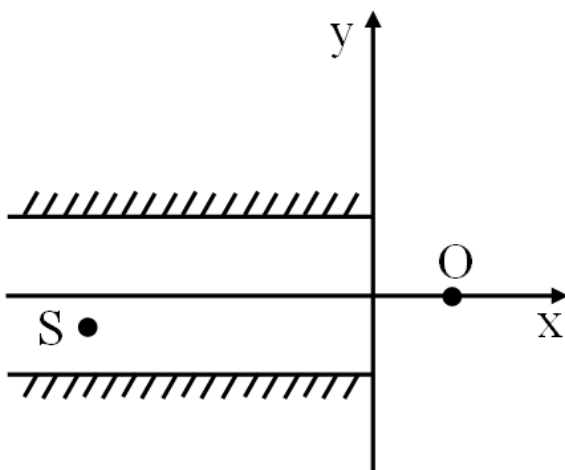
2) Two plane mirrors are parallel and separated by a distance of 3.0 cm. As shown in the figure, the mirrors are parallel to the x-axis, equally spaced from it, and perpendicular to the y-axis. A point source S and an observer O both lie in the xy-plane with coordinates :

$$x_s = -6.0 \text{ cm}, y_s = -0.5 \text{ cm}$$

$$x_o = +2.0 \text{ cm}, y_o = 0.0 \text{ cm}$$

Each mirror extends indefinitely in the negative x-direction and terminates at the y-axis; that is, the mirror occupy  $x \leq 0$ .

How many images of the point source S can the observer O see in the mirrors ?



- (A) 3  
 (B) 4  
 (C) 5  
 (D)  $\infty$

3) A resonance column experiment is performed using a tuning fork of frequency 512 Hz. The first and second resonance positions are observed at 16.2 cm and 49.0 cm, respectively. Each resonance position has a possible reading error of  $\pm 0.1$  cm. The tuning-fork frequency has a possible error of  $\pm 2$  Hz. Using the difference between successive resonance lengths to determine the speed of sound, the maximum percentage error in the calculated speed of sound is :

- (A) 0.39%
- (B) 0.61%
- (C) 1.00%
- (D) 1.22%

4) A graded-index optical fibre has a core of radius  $a$ , whose refractive index varies continuously

with radial distance  $r$  from the axis according to  $n^2(r) = \frac{9}{4} - \frac{1}{2} \frac{r^2}{a^2}$ ,  $0 \leq r \leq a$ . The cladding has

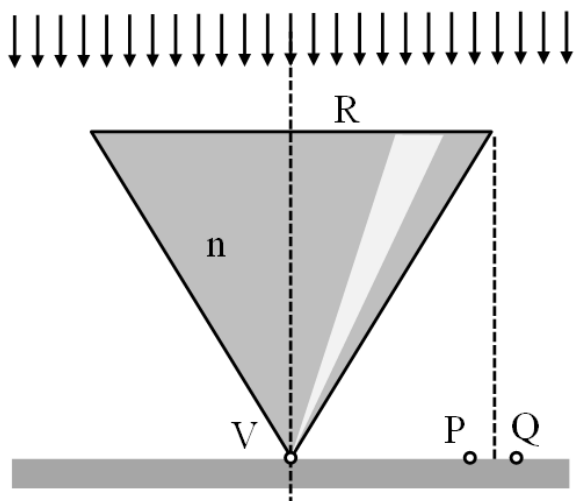
refractive index  $n_c = \frac{\sqrt{7}}{2}$ . A ray enters the fibre through its plane end face from air ( $n^0 = 1$ ) at the axis, making an angle  $30^\circ$  with the fibre axis. The refractive-index profile is independent of the axial coordinate. The ray propagates through the graded-index core and reaches a maximum radial

distance  $r_{\max}$  before bending back toward the axis. Find  $\frac{r_{\max}}{a}$ .

- (A)  $\frac{1}{2}$
- (B)  $\frac{1}{\sqrt{2}}$
- (C)  $\frac{\sqrt{3}}{2}$
- (D) 1

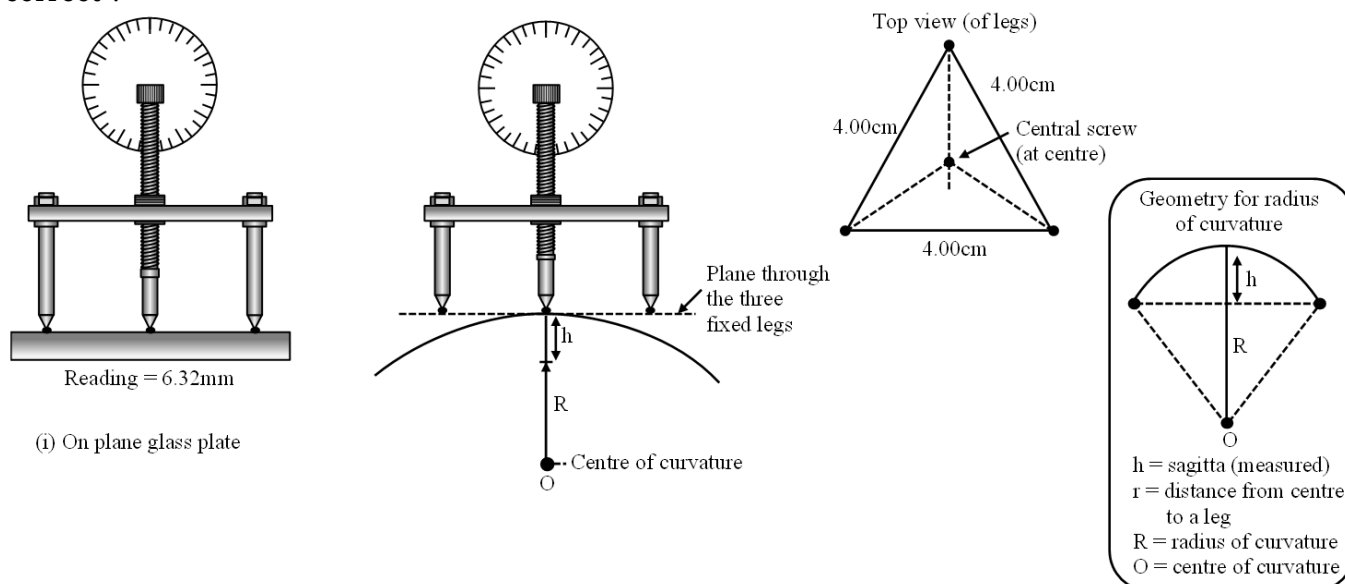
#### SECTION-I (ii)

1) A perfectly transparent glass cone of refractive index  $n = 1.5$  and base radius  $R$  has an axial cross-section in the form of an equilateral triangle. The cone is placed with its axis vertical and vertex  $V$  touching a horizontal plane. A broad uniform beam of monochromatic light consisting of vertically downward parallel rays illuminates the cone and the surrounding plane. The intensity of the incident beam is  $I_0$ . Neglect absorption at all surfaces. Let  $\rho$  denote the radial distance of a point on the horizontal plane from  $V$ . Which of the following statements are correct ?



- (A) A ray entering the horizontal base at radial distance  $r$  from the axis reaches the horizontal plane at  $\rho = 2r$ .
- (B) The intensity on the plane due only to rays entering through the base is  $I_{\text{cone}} = \frac{I_0}{4}$ ,  $0 < \rho < 2R$ .
- (C) The total intensity immediately inside the circle  $\rho = R$  is  $\frac{I_0}{4}$  while immediately outside it is  $\frac{5I_0}{4}$ .
- (D) If  $P$  and  $Q$  lie infinitesimally inside and outside the circle  $\rho = R$ , respectively, then  $\frac{I(P)}{I(Q)} = \frac{1}{3}$ .

2) A spherometer has three fixed legs at the vertices of an equilateral triangle of side 4.0 cm. The central screw has a pitch of 0.50 mm and its circular scale has 50 divisions. When placed on a plane glass plate, the reading is 6.32 mm. When placed on a convex spherical surface, the reading is 5.76 mm. The side of the triangle is measured with a maximum error of 0.01 cm, and each spherometer reading has a maximum error equal to one least count. Which of the following statements are correct ?



- (A) The sagitta of the spherical surface is  $h = 0.56$  mm with maximum error 0.02 mm.
- (B) The distance of the central screw from each fixed leg is  $\frac{4}{\sqrt{3}}$  cm.
- (C) The radius of curvature of the spherical surface is approximately 47.6 cm.

$\frac{h}{2}$

- (D) Neglecting the  $\frac{h}{2}$  term in the expression for radius of curvature, the maximum percentage error in the calculated radius is approximately 4.1%.

3) Three capacitors and three inductors are measured as

$$C_1 = (4.00 \pm 0.08) \mu\text{F}, C_2 = (6.00 \pm 0.06) \mu\text{F}, C_3 = (3.00 \pm 0.03) \mu\text{F},$$

$$L_1 = (8.00 \pm 0.16) \text{mH}, L_2 = (4.00 \pm 0.04) \text{mH}, L_3 = (6.00 \pm 0.06) \text{mH}.$$

All uncertainties represent maximum possible errors. Neglect mutual inductance. Which of the following statements are correct ?

- (A) If  $C_1$  and  $C_2$  are connected in series and this combination is connected in parallel with  $C_3$ , the equivalent capacitance is  $5.40 \mu\text{F}$  with maximum percentage error approximately 1.67%.
- If  $C_1$  and  $C_2$  are connected in parallel and this combination is connected in series with  $C_3$ , the
- (B) equivalent capacitance is approximately  $2.31 \mu\text{F}$  with maximum percentage error approximately 1.09%.
- If  $L_1$  and  $L_2$  are connected in parallel and this combination is connected in series with  $L_3$ , the
- (C) equivalent inductance is approximately  $8.67 \text{mH}$  with maximum percentage error approximately 1.10%.
- If  $L_1$  and  $L_2$  are connected in series and this combination is connected in parallel with  $L_3$ , the
- (D) equivalent inductance is  $4.00 \text{mH}$  with maximum percentage error approximately 1.67%.

4) A prism of apex angle  $A = 45^\circ$  and refractive index  $\mu = \sqrt{2}$  is placed in air. For all rays travelling in the principal section and emerging through the second face, the deviation  $\delta$  is plotted against the angle of incidence  $i$ . Let P and Q be two distinct points on the  $\delta - i$  curve corresponding to the same deviation  $\delta = 30^\circ$ , with incidence angles  $i_1 < i_2$ . Which of the following statements are correct ?

- (A) The complete  $\delta - i$  curve extends from  $i = 0^\circ$  to  $i = 90^\circ$ , and the deviation at both limiting ends is  $45^\circ$ .
- (B) 
$$i = e = \sin^{-1} \left( \sqrt{1 - \frac{1}{\sqrt{2}}} \right), \text{ and } \delta_{\min} = 2\sin^{-1} \left( \sqrt{1 - \frac{1}{\sqrt{2}}} \right) - 45^\circ.$$
  
At the minimum of the curve,
- (C) For the two points P and Q,  $i_1 + i_2 = 60^\circ$ .
- (D) If the slopes of the  $\delta - i$  curve at P and Q are  $m_1$  and  $m_2$  respectively, then  $(m_1 - 1)(m_2 - 1) = 1$ .

5) A thin uniformly charged spherical shell of radius  $R = (100 \pm 0.1) \text{cm}$  carries charge  $Q = (6.00 \pm 0.06) \mu\text{C}$  and rotates uniformly about a diameter. The time for 50 revolutions is measured as  $t = (100.0 \pm 0.4) \text{s}$ . The magnetic field is measured at an axial point at distance  $r_a = (400 \pm 2) \text{cm}$  and at an equatorial point at distance  $r_e = (300 \pm 2) \text{cm}$ , both measured from the centre of the shell. Treat the rotating shell as a magnetic dipole and assume all quoted uncertainties are maximum possible errors. Which of the following statements are correct ?

- (A) The maximum percentage error in the magnetic dipole moment of the rotating shell is approximately 3.4%.
- (B) The maximum percentage error in the magnetic field at the axial point is approximately 4.9%.
- (C) The maximum percentage error in the magnetic field at the equatorial point is approximately 6.4%.
- (D) Although the axial and equatorial magnetic fields have different numerical values, their percentage errors would be equal if  $r_a$  and  $r_e$  had the same percentage uncertainty.

6) A thin semi-transparent spherical mirror is polished on both sides. Its radius of curvature is 10 m. Ram and Shyam stand on opposite sides of the mirror along its principal axis, with a separation of 6 m. Ram observes that the upright diminished image of himself formed by the mirror coincides exactly in position and size with Shyam, whom he can see through the semi-transparent mirror. Ram's height is 180 cm. Which of the following statements are correct? Assume paraxial rays.

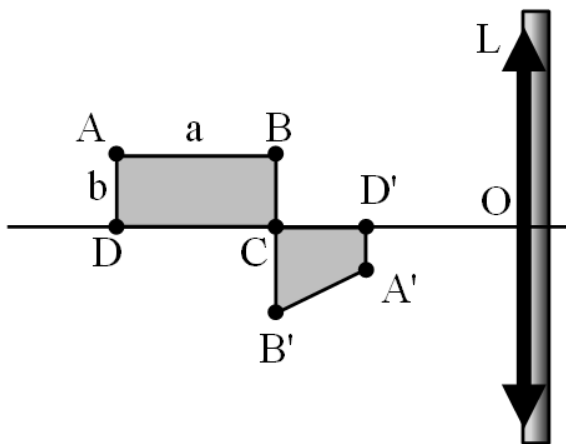
- (A) Ram is on the convex side of the mirror, while Shyam is on the concave side.
- (B) Ram and Shyam are at distances  $(\sqrt{34} - 2)\text{m}$  and  $(8 - \sqrt{34})\text{m}$ , respectively, from the mirror.
- (C) Shyam's height is approximately 102 cm.
- (D) When Shyam looks toward the mirror, he sees an upright enlarged image of himself that coincides in position and size with Ram.

### SECTION-III

1) Two thin converging lenses  $L_1$  and  $L_2$ , of focal lengths 20 cm and 10 cm, are separated by 30cm. An object is placed 45cm to the left of  $L_1$ . A plane-parallel glass slab of thickness 6 cm and refractive index 1.50 is placed between the lenses. A concave mirror of focal length 5 cm is placed to the right of  $L_2$ , facing the lenses. The distance  $D$  between  $L_2$  and the mirror is adjusted so that, after reflection and the return passage through both lenses and the slab, the rays finally emerging to the left of  $L_1$

are parallel. Find  $\frac{D}{2}$  (in cm).

2) A very thin plano-convex lens of refractive index  $n = \frac{3}{2}$  has its plane surface attached to an ideal plane mirror. The radius of curvature of its convex surface is 24 cm. The system is in air. A rectangular object ABCD is placed in front of the lens such that CD lies along the principal optical axis,  $AB \parallel CD$ , and  $CD = 24$  cm,  $BC = 8$  cm. The end C is positioned such that, after refraction through the lens, reflection from the plane mirror, and refraction through the lens again, the point C is imaged exactly onto itself. The complete image of the rectangle is a right trapezoid. Let  $R = \frac{\text{area of the image trapezoid}}{\text{area of the object rectangle}}$ . Neglect the separation between the lens and mirror and work in the paraxial approximation. Find the numerical value of  $8R$ .



3) A modified vernier callipers has 20 vernier-scale divisions equal to 19 main-scale divisions. One main-scale division is 1 mm. When the jaws are fully closed, the zero of the vernier lies 0.30 mm to the right of the main-scale zero. A cylindrical rod is placed between the jaws. The vernier zero lies

between 24 mm and 25 mm of the main scale, and the 8<sup>th</sup> vernier division coincides with a main-scale mark. If the corrected diameter of the rod is x mm, enter the integer value of 10 (x - 24).

4) A concave lens forms a virtual image of a real object. Using the Cartesian sign convention, its v-u curve passes through the point P (-60cm, -15 cm). The tangent to the v-u curve at P intersects the u-axis at A and the v-axis at B. If the coordinates of these intercepts are (u<sub>A</sub>, 0) and (0, v<sub>B</sub>), respectively, find |u<sub>A</sub>v<sub>B</sub>| in cm<sup>2</sup>. Fill the value of  $\frac{\sqrt{|u_A v_B|}}{9}$ .

5) A small object is placed in a liquid of refractive index  $\frac{4}{3}$  in front of a convex spherical surface separating the liquid from a transparent medium of refractive index 2. The centre of curvature of the surface lies in the second medium, and its radius of curvature is 15 cm. The object is situated 35 cm from the centre of curvature of the surface along the principal axis. A paraxial image is formed due to refraction at the spherical surface. If the magnitude of the transverse magnification is m, find the integer value of m.

6) A light ray propagates in a non-homogeneous medium whose refractive index varies with vertical coordinate y according to  $n(y) = \sqrt{2 + \frac{y}{4}}$ , where y is measured in cm. The refractive index is independent of x. At y = 0, a light ray starts from the origin making an angle 45° with the positive x-axis. The ray bends continuously as it travels through the medium. When the ray reaches the level y = 12 cm, find its horizontal displacement x, in cm, from the starting point.

7) Consider the following experimentally reported values :

(1) 0.004050 (2) 4050 (3)  $4.050 \times 10^3$  (4) 0.04050 (5)  $4.05 \times 10^{-3}$  (6) 4005 (7) 0.0040050 (8) 40.50 (9) 4.000 (10) 0.000400

How many of the above quantities contain exactly four significant digits ?

8) In an experiment to determine the resistance of a conductor using Ohm's law, the corrected potential difference and current are  $V = k_v(V_2 - V_1)$ ,  $I = k_i(I_2 - I_1)$ , where  $k_v = 1.00 \pm 0.01$  and  $k_i = 1.00 \pm 0.01$ . The voltmeter readings are  $V_1 = (2.40 \pm 0.03)$  V and  $V_2 = (8.40 \pm 0.03)$  V, while the ammeter readings are  $I_1 = (0.150 \pm 0.005)$  A and  $I_2 = (0.650 \pm 0.005)$  A. If  $R = \frac{V}{I}$ , determine the maximum percentage error in the calculated resistance.

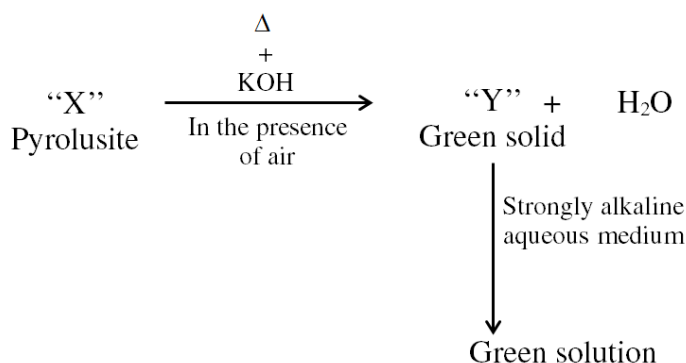
## PART-3 : CHEMISTRY

### SECTION-I (i)

1) Ba<sup>2+</sup>, Sr<sup>2+</sup> and Ca<sup>2+</sup> belong to Group-V cationic analysis.

Select the **CORRECT** statement regarding their analytical properties.

- (A) In the flame test,  $\text{Ba}^{2+}$ ,  $\text{Sr}^{2+}$  and  $\text{Ca}^{2+}$  give grassy-green, brick-red, and crimson-red flames respectively.
- (B) Oxalates of all three cations are white in colour as well as insoluble in acetic solution.
- (C) Sulphate of all three cations are white in colour and also soluble in dilute  $\text{HCl}$  as well as dilute  $\text{HNO}_3$ .
- (D) All three cations can be separated by a yellow solution of chromate followed by addition of aqueous  $\text{CH}_3\text{COOH}$ .



- 2) Select the **INCORRECT** information.
- (A) Paramagnetic tetrahedral potassium salt of green solution is isomorphous with potassium salt of chromate solution.
- In acidic medium green solution show a disproportionation reaction having n-factor is equal to
- (B)  $\frac{2}{3}$
- (C) Potassium salt "Y" in the presence of ozone gives all diamagnetic products.
- (D) Reaction becomes more easier for the conversion of "X" to "Y" in the presence of  $\text{KClO}_3$  as compared to air.

3) Select the **INCORRECT** option :

- (A)  $\text{AgCl}$ ,  $\text{AgBr}$ , and  $\text{AgI}$  are soluble in excess hypo solution, but in excess  $\text{NH}_4\text{OH}$  solution only  $\text{AgCl}$  dissolves completely and readily.
- (B)  $\text{AgCl}$  does not give the chromyl chloride test due to its less dissociation.
- (C) White  $\text{Pb}(\text{OH})_2$  does not dissolve in excess  $\text{NH}_3$  but dissolves in excess  $\text{NaOH}$  solution.
- (D)  $\text{Hg}(\text{NO}_3)_2$  gives a black ppt of  $\text{HgS}$  with  $(\text{NH}_4)_2\text{S}$ , which is dissolved in concentrated  $\text{HNO}_3$  solution.

4) **(P)** is one of the non-electrolytic complex compound and has least Crystal Field Splitting Energy (CFSE) in a series of complex compounds produced on reaction of cobalt(III) chloride with ammonia (in various condition). In another yellow complex compound of the series **(Q)** all chloride ions are bonded to metal via primary valency.

Then select **INCORRECT** information related with series of complex compounds produced on reaction of cobalt(III) chloride with ammonia (in various condition).

- (A) Aqueous solution of complex **(P)** has higher freezing point as compared to the aqueous solution of yellow form **(Q)** of same molality.
- (B) **(P)** can exist in two different colour and they are actually stereoisomers of each other.

- (C) Yellow compound (**Q**) having maximum conductivity and has crystal field stabilization energy equal to  $-2.4 \Delta_0 + 2 \text{ P.E.}$  (P.E. = Pairing energy).
- (D) All complex compounds of series are ionization isomers of each other.

#### SECTION-I (ii)

1) A species "**X**" which is red in colour, in aqueous solution it gives species "**Y**" having purple/violet colouration with species "**Z**" (having rotten egg smell in gaseous form) in an alkaline medium. Select the **CORRECT** statement(s) regarding the given information :

- (A) Both species "**X**" & "**Y**" have same number of linkage isomers.
- (B) Both "**X**" & "**Y**" are inner orbital paramagnetic low spin complex species.
- (C) "**Z**" can give black ppt. with  $\text{K}_3[\text{Cu}(\text{CN})_4]$ .
- (D) "**Z**" gives blue colouration in starch iodate solution

2) Both  $\text{KMnO}_4$  and  $\text{K}_2\text{Cr}_2\text{O}_7$  can be used as a strong oxidising agent in laboratory in redox titrations. Select the **INCORRECT** option(s) :

- (A) On heating crystal of both give at least one green colour product with a paramagnetic gas.
- (B) Both can be used as a primary standard in the laboratory.
- (C) Aqueous solution of both show redox reaction with aqueous HCl.
- (D) All Metal-Oxygen bond lengths are identical within each ion i.e. in  $\text{MnO}_4^-$  and  $\text{Cr}_2\text{O}_7^{2-}$ .

3) Anions  $\text{X}^-$ ,  $\text{Y}^-$  and  $\text{Z}^{2-}$  containing "Carbon" can be differentiated by concentrated  $\text{H}_2\text{SO}_4$  but not with a neutral solution of  $\text{FeCl}_3$ .

$\text{Y}^-$  gives a silver mirror test with ammoniacal  $\text{AgNO}_3$  solution, but none of the anion produce  $\text{CO}_2$  with dilute  $\text{H}_2\text{SO}_4$ .

Select the **CORRECT** statement(s) :

- (A)  $\text{Z}^{2-}$  with dilute sulphuric acid, there is no visible action but in the presence of manganese dioxide carbon dioxide is evolved.
- (B)  $\text{X}^0$  can be confirmed by the Cannizzaro reaction.
- (C) In acidic medium  $\text{Z}^{2-}$  reacts with  $\text{KMnO}_4$  solution and reaction is catalyzed by  $\text{Mn(II)}$  ion.
- (D)  $\text{Y}^-$  and  $\text{Z}^{2-}$  both produce  $\text{CO}_2$  gas with conc.  $\text{H}_2\text{SO}_4$ .

4) In the analysis of Group III cations ( $\text{A}^{+3}$ ,  $\text{B}^{+3}$  &  $\text{C}^{+3}$ )  $\text{NH}_4\text{OH}$  with  $\text{NH}_4\text{Cl}$  solution is used as a group reagent.

Then select the **CORRECT** option(s).

- (A) If  $\text{A}^{3+}$  can be confirmed by a blood-red colouration with KSCN, then metal "**A**" has the 2<sup>nd</sup> highest value of ionization energy among the elements of the 3d-series.
- (B) If  $\text{B(OH)}_3$  is a green ppt. which is soluble in excess NaOH but partially soluble in excess  $\text{NH}_4\text{OH}$  solution, then the metal "**B**" has acidic oxide in +6 oxidation state
- (C) If  $\text{C}_2\text{O}_3$  is an amphoteric oxide and  $\text{C(OH)}_3$  is a gelatinous white ppt, then  $\text{C}_2\text{O}_3$  is major component of Emerald gem stone.

- (D) Gelatinous  $\text{C}(\text{OH})_3$  adsorbs an organic dye like blue litmus to form a floating blue-colored complex called a "Lake".

5) Select the **CORRECT** option(s).

- (A) Sodium (ethylenediaminetetraacetato)nickelate(II) can show geometrical as well as optical isomerism.
- (B) Aquated  $\text{TiCl}_4$  is colourless and  $\text{TiCl}_4$  solid with  $\text{Al}(\text{CH}_3)_3$  is basis of the ziegler catalysts.
- (C) Chloride of metal "M" is used as a catalyst in the Deacon process, but iodide of metal "M" in the same oxidation state does not exist.
- (D) Oxidation state of metal is same in cis-platin and complex formed by reaction of platinum and 3 : 1 v/v ratio of conc.  $\text{HCl}$  and conc.  $\text{HNO}_3$ .

6) Select **CORRECT** option(s) about given three complexes of Cobalt -

(I)  $[\text{Co}(\text{bipy})_3]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$

(II)  $\text{K}_2[\text{Co}(\text{CN})_4]$

(III)  $\text{K}_2[\text{Co}(\text{NCS})_4] \cdot 4\text{H}_2\text{O}$

- (A) Only one complex show stereo isomerism.
- (B) Only two complexes have Co-N linkage(s)
- (C) Only two complexes use d-orbital(s) in hybridization.
- (D) None of the complex is diamagnetic & all have different geometry around central metal.

### SECTION-III

1) Number of paramagnetic species having :  $\mu \geq \sqrt{8}$  B.M.

$[\text{Fe}(\text{NH}_3)_6]^{3+}$ ,  $[\text{Ni}(\text{CO})_4]$ ,  $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ ,  $\text{O}_2[\text{AsF}_6]$ ,  $[\text{Cr}(\text{NH}_3)_6]^{2+}$ ,  $[\text{NiF}_6]^{2-}$ ,  $[\text{Ag}(\text{S}_2\text{O}_3)]^{-1}$ ,  $[\text{Ni}(\text{en})_3]\text{S}_2\text{O}_3$ ,  $[\text{Fe}(\text{NH}_3)_6]^{2+}$

2) Number of **CORRECT** order.

- (1)  $\text{Trans-}[\text{Cu}(\text{en})_2(\text{H}_2\text{O})_2]^{2+} > \text{Cis-}[\text{Cu}(\text{en})_2(\text{H}_2\text{O})_2]^{2+}$  : Stability order
- (2)  $[\text{Mn}_2(\text{CO})_{10}] > [\text{Mn}(\text{CO})_5]$  : Oxidising power
- (3)  $\text{K}_3[\text{Fe}(\text{CN})_6] > \text{K}_4[\text{Fe}(\text{CN})_6]$  : Crystal field splitting energy  $\Delta_o$
- (4)  $[\text{Co}(\text{gly})_3] > [\text{Co}(\text{en})_3]^{3+}$  : Number of optically inactive isomers
- (5)  $[\text{PtCl}_2\text{Br}_2]^{2-} > [\text{Zn}(\text{gly})_2]$  : Number of total stereoisomers

3) Thermal decomposition is a process where a single substance breaks down into two or more substances. then number of **CORRECT** heating processes :

- (A) Green Vitriol  $\xrightarrow[\text{temperature}]{\text{high}}$  Metal oxide with all gaseous/volatile product(s) (at reaction condition) having polar as well as planar nature .
- (B)  $\text{Cu}(\text{NO}_3)_2 \xrightarrow{250^\circ\text{C}}$  Metal oxides with paramagnetic gaseous product(s).
- (C)  $\text{Na}_2\text{SO}_3 \cdot 5\text{H}_2\text{O} \xrightarrow{\text{Strongly Heated}}$  at least one product(s) has S-S linkage.
- (D)  $\text{NH}_4\text{ClO}_4 \xrightarrow{\Delta}$  all diamagnetic product(s).
- (E) White Lead  $\xrightarrow{\Delta, (350^\circ\text{C})}$  at least three product(s) are gaseous/volatile (at reaction condition) in

nature.

4) Number of colored compounds (excluding white) :

$\text{Bi}_2\text{S}_3$  ,  $[\text{Ni}(\text{NH}_3)_6]^{2+}$  ,  $\text{Sb}_2\text{S}_3$  ,  $\text{SnS}_2$  ,  $\text{K}_2\text{Ca}[\text{Fe}(\text{CN})_6]$ ,  $\text{CuSCN}$ ,  $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$ ,  $\text{Bi}(\text{OH})_3$  ,  $\text{Mg}_2\text{P}_2\text{O}_7$  ,  $\text{PbO}_2$

5) Consider the following conditions :

(A) Paramagnetic as well as inner orbital complex species.

(B) Can show linkage as well as geometrical isomerism.

(C) Can coordination as well as optical activity.

(D) Can show optical but not geometrical isomerism.

Number of correct species which follow atleast two conditions :

(1)  $[\text{Pt}(\text{en})_2\text{Cl}_2]$   $[\text{PtCl}_2\text{Br}_2]$

(2)  $[\text{Cr}(\text{en})_3]$   $[\text{Cr}(\text{ox})_3]$

(3)  $[\text{Ag}(\text{NH}_3)_2]$   $[\text{Ag}(\text{CN})_2]$

(4)  $[\text{Fe}(\text{NH}_3)_6]$   $[\text{Fe}(\text{CN})_4(\text{NO}_2)_2]$

6) Number of **CORRECT** order.

(1)  $\text{CrO}_3 > \text{MoO}_3 > \text{WO}_3$  : Oxidising power

(2) 5d series > 4d series : Density

(3)  $\text{Cu}^{2+}/\text{Cu} > \text{Zn}^{2+}/\text{Zn}$  : Negative value of Standard electrode potential  $[E_{\text{M}^{+2}/\text{M}}^\ominus]$  in volt

(4)  $\text{Na}_2\text{Cr}_2\text{O}_7 > \text{K}_2\text{Cr}_2\text{O}_7$  : Deliquescent nature

(5)  $d^7$  (octahedral, low spin) >  $d^5$  (octahedral, high spin) : Jahn-Teller Distortion

(6)  $\text{V} > \text{Zn}$  : Enthalpy of atomization

7) 17.4 gm pyrolusite ore is react with excess of HCl solution. The liberated gas is passed through excess of KI solution. If the yellow brown solution obtained exactly requires 40 ml of decimolar hypo solution for titration, calculate the percentage of pure  $\text{MnO}_2$  in the pyrolusite ore.

Atomic mass of O = 16, Cl = 35.5, H = 1, K = 39, I = 127, Mn = 55, Cr = 52, Pb = 207.2, Fe = 56.

8) Consider different complexes where  $\text{M}^{+2}$  represent  $\text{Ni}^{+2}$  (X, Y are neutral monodentate ligands) then number of correct statements .

(1) If  $[\text{MX}_2\text{Y}_2]^{2+}$  is paramagnetic then it has planar geometry at metal

(2) If  $[\text{MX}_2\text{Y}_2]^{2+}$  is diamagnetic then it has two stereoisomers

(3)  $[\text{MX}_6]^{+2}$  is paramagnetic with  $\text{sp}^3\text{d}^2$  hybridisation

(4) Number of valence electrons on metal  $[\text{MX}_4]^{+2}$  and  $[\text{MY}_6]^{+2}$  are same

(5) Aquated  $\text{M}^{+2}$  is green in colour.

## ANSWER KEYS

### PART-1 : MATHEMATICS

#### SECTION-I (i)

| Q. | 1 | 2 | 3 | 4 |
|----|---|---|---|---|
| A. | B | A | B | A |

#### SECTION-I (ii)

| Q. | 5   | 6       | 7     | 8   | 9   | 10 |
|----|-----|---------|-------|-----|-----|----|
| A. | B,D | A,B,C,D | A,B,C | A,C | A,D | B  |

#### SECTION-III

| Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|----|----|----|----|----|----|----|----|----|
| A. | 2  | 2  | 7  | 3  | 4  | 3  | 4  | 7  |

### PART-2 : PHYSICS

#### SECTION-I (i)

| Q. | 19 | 20 | 21 | 22 |
|----|----|----|----|----|
| A. | A  | A  | C  | B  |

#### SECTION-I (ii)

| Q. | 23    | 24      | 25  | 26    | 27 | 28      |
|----|-------|---------|-----|-------|----|---------|
| A. | A,B,C | A,B,C,D | B,C | A,B,D | D  | A,B,C,D |

#### SECTION-III

| Q. | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 |
|----|----|----|----|----|----|----|----|----|
| A. | 5  | 3  | 1  | 5  | 3  | 8  | 6  | 5  |

### PART-3 : CHEMISTRY

#### SECTION-I (i)

| Q. | 37 | 38 | 39 | 40 |
|----|----|----|----|----|
| A. | D  | C  | D  | D  |

#### SECTION-I (ii)

| Q. | 41 | 42    | 43      | 44      | 45  | 46      |
|----|----|-------|---------|---------|-----|---------|
| A. | D  | B,C,D | A,B,C,D | A,B,C,D | B,C | A,B,C,D |

SECTION-III

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| Q. | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 |
| A. | 3  | 2  | 3  | 5  | 2  | 6  | 1  | 3  |

## SOLUTIONS

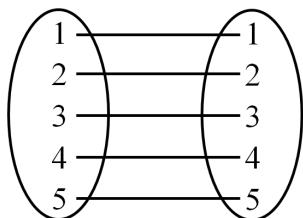
### PART-1 : MATHEMATICS

$$1) 2^9 - (1 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9) \\ = 512 - 46 = 466$$

$$2) x = y = 1 \\ (f(1))^2 - f(1) - 2 = 0 \\ \Rightarrow (f(1) - 2)(f(1) + 1) = 0 \\ \Rightarrow f(1) = -1, 2 \\ x = y = 0 \\ (f(0))^2 - f(0) = 0 \Rightarrow f(0) = 0, 1 \\ x = 0, y = 1 \\ f(0)f(1) - f(0) = 1 \\ f(0)(f(1) - 1) = 1 \\ \text{Here } f(0) = 0 \text{ is not possible} \\ \square f(0) = 1 \text{ and } f(1) = 2 \\ \text{Now, } y = 1 \\ f(x).2 - f(x) = x + 1 \\ \Rightarrow f(x) = x + 1 \\ \square f(31) = 32$$

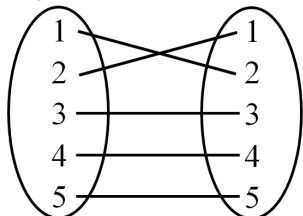
$$3) \text{ Let } \cos^{-1} \frac{2}{\sqrt{7}} = \theta \Rightarrow \cos \theta = \frac{2}{\sqrt{7}} \quad \theta \in \left(0, \frac{\pi}{4}\right) \\ \sin \theta = \frac{\sqrt{3}}{\sqrt{7}} \Rightarrow \sin 2\theta = 2 \cdot \frac{\sqrt{3}}{\sqrt{7}} \cdot \frac{2}{\sqrt{7}} = \frac{4\sqrt{3}}{7} \\ \tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \frac{3 \cdot \frac{\sqrt{3}}{2} - \frac{3\sqrt{3}}{8}}{1 - 3 \cdot \frac{3}{4}} = -\frac{9\sqrt{3}}{10} \\ \tan(\pi - 3\theta) = \frac{9\sqrt{3}}{10} \\ \frac{3}{2} \cos^{-1} \left( \frac{2}{\sqrt{7}} \right) + \frac{1}{2} \sin^{-1} \left( \frac{4\sqrt{3}}{7} \right) + \frac{5}{6} \tan^{-1} \left( \frac{9\sqrt{3}}{10} \right) \\ = \frac{3}{2} \theta + \frac{1}{2} \sin^{-1} (\sin 2\theta) + \frac{5}{6} \tan^{-1} (\tan (\pi - 3\theta)) \\ = \frac{3}{2} \theta + \frac{1}{2} \cdot 2\theta + \frac{5}{6} (\pi - 3\theta) \\ = \frac{5\pi}{6}$$

$$4) \text{ Case-I : } f(i) = i \quad \forall i \in A = 1$$

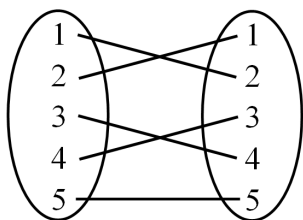


Case-II :  $f(i) = i$  for exactly 3 elements but one pair interchanged.

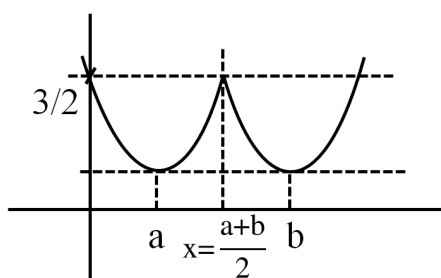
$${}^5C_3 \times 1 = 10$$



Case-III :  $f(i) = i$  for exactly one element but two pair of elements interchanged  ${}^5C_1 \times 3 = 15$



□ Total no. of functions =  $1 + 10 + 15 = 26$ .



5)

$$f(x) = \min\{(x-a)^2 + 2a^2, (x-b)^2 + 2a^2\}$$

$$2a^2 = 1 \Rightarrow a^2 = \frac{1}{2}$$

$$(x-a)^2 + 2a^2 = (x-b)^2 + 2a^2$$

$$x-a = -x+b \Rightarrow x = \frac{a+b}{2}$$

$$f\left(\frac{a+b}{2}\right) = \frac{(b-a)^2}{4} + 2a^2$$

$$\text{for } f(x) = \frac{3}{2} \text{ to have 3 solutions, } \frac{(b-a)^2}{4} + 2a^2 = 3a^2$$

$$\text{for } f(x) = \frac{3}{2} \text{ to have 7 solutions } \frac{(b-a)^2}{4} = a^2$$

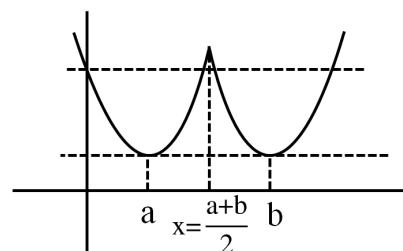
$$\Rightarrow b-a = 2a$$

$$b = 3a$$

$$b = \frac{3}{\sqrt{2}}$$

$$\frac{(b-a)^2}{4} + 2a^2 > 3a^2$$

$$\frac{b-a}{2} > a \Rightarrow b > 3a$$



$$b > \frac{3}{\sqrt{2}}$$

□ Least integral value of b is 3

$$6) \tan^{-1}\{x\} - \cot^{-1}[x] > 0$$

x can not be negative

$$\text{C-I : } x \in [0, 1) \Rightarrow \tan^{-1}x > \frac{\pi}{2} \Rightarrow \text{No solution}$$

$$\text{C-II : } x \in [1, 2) \Rightarrow \tan^{-1}(x-1) > \tan^{-1}1 \Rightarrow x > 2 \Rightarrow \text{No solution}$$

$$\text{C-III : } x \in [2, 3) \Rightarrow \tan^{-1}(x-2) > \tan^{-1}\frac{1}{2} \Rightarrow x > 2 + \frac{1}{2}$$

$$\square x \in \left(2 + \frac{1}{2}, 3\right)$$

$$\text{C-IV : } x \in [n, n+1), n \in \mathbb{N}, n \geq 2$$

$$\tan^{-1}(x-n) > \tan^{-1}\frac{1}{n} \Rightarrow x > n + \frac{1}{n}$$

$$\square D_f : \left(n + \frac{1}{n}, n+1\right), n \in \mathbb{N}, n \geq 2$$

**For range :**

$$f(x) = \tan^{-1}\{x\} - \cot^{-1}[x]$$

$$x \in \left(n + \frac{1}{n}, n+1\right), n \in \mathbb{N}, n \geq 2$$

$$f(x) = \ln \left( \tan^{-1}(x-n) - \tan^{-1}\frac{1}{n} \right)$$

$$x \rightarrow n+1, f(x) \rightarrow \ln \left( \tan^{-1}1 - \tan^{-1}\frac{1}{n} \right) = \ln \left( \tan^{-1} \left( \frac{n-1}{1+n} \right) \right)$$

$$\square \text{Range is } \left(-\infty, \ln \tan^{-1} \left( \frac{n-1}{n+1} \right) \right), n \in \mathbb{N}, n \geq 2$$

$$7) (A) n((A \times B) \cap (B \times A)) = n(A \cap B) \cdot n(B \cap A) = r^2$$

$$(B) n((A \times B) \cup (B \times A)) = n(A \times B) + n(B \times A) - n((A \times B) \cap (B \times A)) \\ = 2pq - r^2$$

$$(C) n((A \times A) \cap (B \times B)) = n(A \cap B) \cdot n(B \cap A) = r^2$$

$$(D) n((A \times A) \cup (B \times B)) = n(A \times A) + n(B \times B) - n((A \times A) \cap (B \times B)) \\ = p^2 + q^2 - r^2$$

$$8) (A) g(x) = [x] \{x\} - [-x] \{-x\}$$

$$\text{Case-I : } x \in \mathbb{I}$$

$$g(x) = 0 \Rightarrow '0' \text{ is in the range}$$

$$\text{Case-II : } x \notin \mathbb{I}, \text{ let } x = n + f \quad f \in (0, 1)$$

$$g(x) = nf - (-n-1)(-n-f-(-n-1))$$

$$= nf + (n+1)(-n-f+n+1)$$

$$= nf - nf + n - f + 1$$

$$= n - f + 1 \notin \mathbb{I}$$

□ '0' is only the integer in the range of g(x).

$$(C) g(x) = x$$

$$\text{Case-I : } x \in \mathbb{I} \Rightarrow g(x) = 0 \Rightarrow x = 0 \text{ is the solution.}$$

$$\text{Case-II : } x \notin I \Rightarrow g(x) = x \Rightarrow n - f + 1 = n + f \Rightarrow f = \frac{1}{2}$$

$$\square \text{ solution of the equations is } = n + \frac{1}{2}, n \in I$$

$$\bigcup \left\{ n + \frac{1}{2} \right\} \cup \{0\}$$

Complete solution set will be,  $n \in I$

$$\square \frac{-9}{2}, \frac{-7}{2}, \frac{-5}{2}, \frac{-3}{2}, \frac{-1}{2}, 0, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}, \dots, \frac{19}{2}$$

$$\text{Reqd. sum} = \frac{1}{2} \left( \frac{11}{2} + \frac{19}{2} \right) \times 5 = \frac{30}{4} \times 5 = \frac{75}{2}$$

$$9) (A) \quad R_f : \left[ \frac{1}{7}, 7 \right] \Rightarrow A = \{1, 2, 3, \dots, 6, 7\}$$

$2 \leq a + b \leq 14 \Rightarrow$  perfect square nos. are 4 & 9

$$R = \{(1, 3), (3, 1), (2, 2), (2, 7), (7, 2), (3, 6), (6, 3), (4, 5), (5, 4)\}$$

$$n(R) = 9$$

$$(D) \quad (1, 1), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (6, 1), (1, 6)$$

$\Rightarrow$  8 pairs are to be added to make it transitive.

10)

$$S_n = \sum_{r=1}^n \left( \tan^{-1} \left( \frac{1}{r^2 + r + 1} \right) + \tan^{-1} (r^2 + r + 1) - \tan^{-1} (r^2 - r + 1) \right) + \tan^{-1} \left( \frac{n+4-n}{1+n(n+4)} \right)$$

$$= \sum_{r=1}^n \tan^{-1} \left( \frac{r - (r^{-1})}{1 + r(r^{-1})} \right) + \tan^{-1} (n+4) - \tan^{-1} n$$

$$= \sum_{r=1}^n \left( \tan^{-1} r - \tan^{-1} (r-1) \right) + \tan^{-1} (n+4) - \tan^{-1} n$$

$$S_n = \tan^{-1} (n+4)$$

$$\Rightarrow \tan(S_n) = n+4 > 2030 \Rightarrow n > 2026$$

$\square$  Least natural no. of  $n$  is 2027

$$11) \text{ equation is of form } |a| + |b| = |a - b|$$

12) The sum of the three numbers is an even number, so at least one of them is even. The only even prime number is 2. Only  $3n - 4$  and  $5n - 3$  can be even. Solving the equations  $3n - 4 = 2$  and  $5n - 3 = 2$  yields  $n = 2$  and  $n = 1$ , respectively. It is trivial to check that  $n = 2$  does make all three given numbers prime.

$$13) f(x).f(x+2) = -1 \Rightarrow f(x) = -\frac{1}{f(x+2)}$$

$$x \rightarrow x+2$$

$$f(x+2).f(x+4) = -1 \Rightarrow f(x+2) = -\frac{1}{f(x+4)}$$

$$f(x) = f(x+4) \Rightarrow f \text{ is periodic with period 4}$$

$$f(1001) = f(1) = f(5) = f(9) = 2$$

$$f(3) = -\frac{1}{2} \Rightarrow f(7) = f(11) = \dots = -\frac{1}{2}$$

$$\begin{aligned} & \sum_{r=1}^{101} f(2r-1) \\ \text{Now, } \sum_{r=1}^{101} f(2r-1) &= f(1) + f(3) + f(5) + \dots + f(201) \\ &= \underbrace{f(1) + f(5) + f(9) + \dots + f(201)}_{51 \text{ terms}} + \underbrace{f(3) + f(7) + \dots + f(199)}_{50 \text{ terms}} \\ &= 51 \times 2 + 50 \left(-\frac{1}{2}\right) \\ &= 102 - 25 = 77 \end{aligned}$$

$$\begin{aligned} 14) R_1 &= \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,4), (2,1), (4,3)\} \\ R_2 &= \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,4), (2,1), (4,3), (2,5), (5,2), (5,1), (1,5)\} \\ R_3 &= \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,4), (2,1), (4,3), (4,5), (5,4), (5,3), (3,5)\} \\ \text{Ans.} &= 3 \end{aligned}$$

$$\begin{aligned} 15) \tan^{-1}(x+1) - \tan^{-1}(x-1) &= \frac{\pi}{4} \\ \frac{x+1 - (x-1)}{1 + (x+1)(x-1)} &= 1 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2} \\ \text{Reqd. sum} &= 1 + 1 + \tan^{-1}\sqrt{2} + \left(1 + 1 - \tan^{-1}\sqrt{2}\right) = 4 \end{aligned}$$

$$\begin{aligned} 16) \sin^{-1}(\sin p) + \cos^{-1}(\cos p) &= \frac{\pi}{3} \\ \text{Case-I : } p \in \left[0, \frac{\pi}{2}\right] &\Rightarrow 2p = \frac{\pi}{3} \Rightarrow p = \frac{\pi}{6} \\ \text{Case-II : } p \in \left[2\pi, \frac{5\pi}{2}\right] &\Rightarrow p - 2\pi + p - 2p = \frac{\pi}{3} \\ 2p &= 4\pi + \frac{\pi}{3} \Rightarrow p = \frac{13\pi}{6} \\ \therefore \alpha + \beta &= \frac{14\pi}{6} \Rightarrow \frac{3(\alpha + \beta)}{7} = \pi \\ \left[\frac{3(\alpha + \beta)}{7}\right] &= 3 \end{aligned}$$

$$\begin{aligned} 17) x^2 - 2x &= t \\ \tan^{-1}|t| + \tan^{-1}\left(\frac{1}{t+2}\right) &= \frac{\pi}{4} \\ \tan^{-1}|t| &= \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{t+2}\right) \\ |t| &= \frac{1 - \frac{1}{t+2}}{1 + \frac{1}{t+2}} \\ \Rightarrow |t|(t+3) &= t+1 \\ \text{Case-I : } t \geq 0 &\Rightarrow t^2 + 2t - 1 = 0 \Rightarrow (t+1)^2 = 2 \end{aligned}$$

$$\Rightarrow t = -1 \pm \sqrt{2}$$

$$\begin{array}{cc} \swarrow & \searrow \\ t = -1 - \sqrt{2} & t = -1 + \sqrt{2} \end{array}$$

rejected

Now,  $x^2 - 2x = -1 + \sqrt{2} \Rightarrow (x - 1)^2 = \sqrt{2} \Rightarrow x = 1 \pm \sqrt[4]{2}$  (2 solutions)

**Case-II :**  $t < 0 \Rightarrow t^2 + 4t + 1 = 0 \Rightarrow (t + 2)^2 = 3 \Rightarrow t = -2 \pm \sqrt{3}$

Now,  $x^2 - 2x = -2 + \sqrt{3} \Rightarrow (x - 1)^2 = -1 + \sqrt{3} \Rightarrow x = 1 \pm \sqrt{\sqrt{3} - 1}$

$x^2 - 2x = -2 - \sqrt{3} \Rightarrow (x - 1)^2 = -1 - \sqrt{3} \Rightarrow$  No solutions

Hence, number of solutions = 4

18) No. of symmetric relations containing (2, 3) but not (2, 4) is  $2^5 \cdot 2^8 = 2^{13}$

No. of symmetric relations which are also reflexive and containing (2, 3) but not (2, 4) is  $2^8$

□ Req'd. no. of relations =  $2^{13} - 2^8 = 2^8 \times 31 = m \cdot 2^n$

□  $m - n = 7$

## PART-2 : PHYSICS

19)

Correct Answer is (A)

20)

Correct Answer is (A)

21)

Correct Answer is (C)

22)

Correct Answer is (B)

23)

Correct Answer is (A,B,C)

24)

Correct Answer is (A,B,C,D)

25)

Correct Answer is (B,C)

26)

Correct Answer is (A,B,D)

27)

Correct Answer is (D)

28)

Correct Answer is (A,B,C,D)

29)

Correct Answer is (5)

30)

Correct Answer is (3)

31)

Correct Answer is (1)

32)

Correct Answer is (5)

33)

Correct Answer is (3)

34)

Correct Answer is (8)

35)

Correct Answer is (6)

36)

Correct Answer is (5)

37) A, B, C are incorrect

38)  $X = \text{MnO}_2$

$Y = \text{MnO}_4^{2-}$

39) (A), (B), (C) are correct

40)  $\text{CoCl}_3 \cdot 6\text{NH}_3$  (Q)

$\text{CoCl}_3 \cdot 5\text{NH}_3$

$\text{CoCl}_3 \cdot 4\text{NH}_3$

$\text{CoCl}_3 \cdot 3\text{NH}_3$  (P)

41)  $X = \text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$

$Y = \text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$

$Z = \text{H}_2\text{S}$

42) only statement 1 is correct

43)  $X^- = \text{CH}_3\text{COO}^-$

$Y^- = \text{HCOO}^-$

$Z^{2-} = \text{C}_2\text{O}_4^{2-}$

44)  $\text{A}^{+3} \rightarrow \text{Fe}^{+3}$

$\text{B}^{+3} \rightarrow \text{Cr}^{+3}$

$\text{C}^{+3} \rightarrow \text{Al}^{+3}$

45) Theory based

46) (I) Octahedral, Paramagnetic

(II) Sq. Planar, Paramagnetic

(III) Tetrahedral, Paramagnetic

47)  $[\text{Cr}(\text{NH}_3)_6]^{+2}$ ,  $[\text{Fe}(\text{NH}_3)_6]^{+2}$ ,  $[\text{Ni}(\text{en})_3]\text{S}_2\text{O}_3$

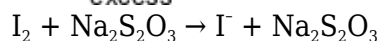
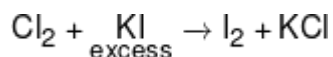
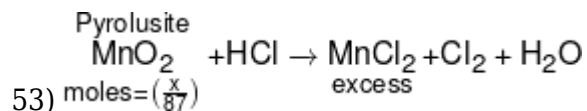
48) Only (1) & (3) are correct

49) (B), (C), (E) are correct

50) First four with  $\text{PbO}_2$  are coloured (excluding white)

51) (2), (4) follow atleast two conditions

52) All are correct



Lets assume pure  $\text{MnO}_2$  in sample is  $x_g$

Eq. of  $\text{MnO}_2 = \text{Eq. of Cl}_2$

$$\frac{n}{87} \times 2 = n(\text{Cl}_2) \times 2 \Rightarrow n(\text{Cl}_2) = \frac{x}{87}$$

Eq. of  $\text{Cl}_2 = \text{Eq. of I}_2$

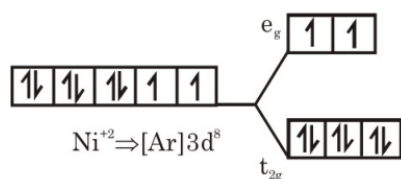
$$\frac{n}{87} \times 2 = 40 \times 10^{-3} \times 0.1 \times 1 = x = 174 \times 10^{-3} \text{ g}$$

mass of  $\text{MnO}_2 = 0.174 \text{ g}$

$$\% \text{ mass of MnO}_2 = \frac{0.174}{17.4} \times 100 = 1\% \text{ in sample}$$

54) (1) If  $[\text{MX}_2\text{Y}_2]^{+2}$  is paramagnetic then it has tetrahedral geometry of metal.

(2) If  $[\text{MX}_2\text{Y}_2]^{+2}$  is diamagnetic then it has two G.I. (cis & trans)



(3)

$[\text{MX}_6]^{+2} \Rightarrow$  paramagnetic &  $sp^3d^2$  hybridisation

(4) Valence  $e^-$  on metal in  $[\text{MX}_4]^{+2} = 16$  and in  $[\text{MY}_6]^{+2} = 20$